Notes: Dimopoulos & Sacchetto (JFE, 2017)
Merger activity in industry equilibrium

Contents

1 Big picture

- **Question.** What is the role of mergers and acquisitions in improving productive efficiency once mergers interact with firms' entry and exit decisions in industry equilibrium?
- **Core mechanism.** Mergers affect productivity directly through realized synergies and indirectly by changing which firms enter and exit the industry.
- **Model.** The paper builds a dynamic competitive industry model with heterogeneous firms, aggregate demand shocks, endogenous search for merger partners, entry, exit, and repeated investment.
- **Calibration.** The model is calibrated to moments on merger rates, merger gains, exit rates, entrant and exiting-firm productivity, and aggregate earnings dynamics.
- **Main quantitative result.** Merger activity raises average firm productivity by about 4.8% relative to a no-merger economy.
- **Direct versus indirect effect.** About 4.1% comes from productivity synergies realized in mergers, while about 0.7% comes from the way merger options change entry and exit decisions.
- **Entry effect.** Because the option to be acquired is valuable, merger activity raises entry; in the calibrated model the entry rate rises from about 5.8% without mergers to 7.9% with mergers.
- **Exit effect.** Because low-productivity firms can hope to be acquired, merger activity lowers exit; in the calibrated model the exit rate falls from about 5.8% without mergers to 3.7% with mergers.
- **Business-cycle result.** Without mergers, booms can be "sullying" because low-productivity incumbents stay and new entrants dilute average productivity. With mergers, procyclical merger synergies reverse much of this pattern.
- **Recession result.** Recessions are cleansing without mergers, but can be sullying when merger options are present because merger synergies become less valuable and fewer low-productivity firms are acquired.
- **Policy result.** A policy that reduces entry costs may not raise average productivity once merger activity is considered. Lower entry costs can reduce prices, weaken merger incentives, reduce merger synergies, and lower average productivity.
- **Validation.** The model predicts negative relations between merger and exit rates, procyclical merger activity when productivity synergies dominate, and lower acquirer returns for large acquirers. The paper documents support for these predictions.

2 Data and measurement

2.1 Industry turnover and merger data

- The empirical objects used for calibration and validation are industry-level merger, entry, exit, productivity, and earnings-growth moments.
- Merger and exit rates are built using CRSP firm delisting information and SDC Platinum merger data.
- Merger gains are measured using combined announcement returns around takeover bids, following evidence summarized in Betton, Eckbo, and Thorburn.
- Entrant and exiting-firm productivity measures use CRSP/Compustat merged data and total factor productivity measures from Imrohoroglu and Tuzel.
- Aggregate earnings-growth moments come from Bureau of Economic Analysis data on the corporate sector.

Interpretation: The calibration targets are chosen to discipline both merger-market activity and the two other sources of industry turnover: entry and exit.

2.2 Core calibration targets

- Average merger rate: model 4.21%, data 4.54%.
- Standard deviation of merger rate: model 0.012, data 0.016.
- Autocorrelation of merger rate: model 0.292, data 0.291.
- Average absolute scaled difference in log Tobin's Q of merged firms: model 0.925, data 0.953.
- Average combined merger gains: model 1.63%, data 1.79%.
- Average exit rate: model 3.71%, data 3.68%.
- Relative TFP of entrants: model 1.060, data 1.229.
- Relative TFP of exiting firms: model 0.722, data 0.791.

Interpretation: The model does not match every target exactly, but it reproduces the key magnitudes needed to study the interaction of mergers, entry, and exit.

2.3 State variables and shocks

The aggregate state vector is approximated as

$$\widehat{x}_t = [s_t, P_t, \log N_t, m_t, \log v_t],$$

where s_t is aggregate demand, P_t is output price, N_t is the mass of incumbents, and m_t and v_t are the cross-sectional mean and variance of log productivity.

Interpretation: This makes the model an industry-equilibrium model: merger decisions depend not only on the productivity of a pair of firms, but also on prices and the full distribution of incumbent productivity.

3 Model and empirical specifications

3.1 Production, demand, and profit

A firm with productivity z and capital k produces

$$q = zk^{\alpha},$$

and sells output in a competitive market with inverse demand

$$P = sQ^{-1/\eta}.$$

Operating profit is

$$\pi(P, z, k) = Pzk^{\alpha} - c_f.$$

Interpretation: The fixed operating cost c_f creates an exit margin and also gives mergers a cost-saving motive.

3.2 Merger productivity technology

When firms i and j merge, the next-period productivity level is

$$\zeta_M(z_i, z_j) = \underline{z} + (\bar{z} - \underline{z}) \left[\lambda \max\{\widetilde{z}_i, \widetilde{z}_j\} + (1-\lambda) \min\{\widetilde{z}_i, \widetilde{z}_j\}\right]^{\theta},$$

where $\widetilde{z} = (z - \underline{z})/(\bar{z} - \underline{z})$, $0 \leq \lambda \leq 1$, and $0 < \theta < 1$.

Interpretation: The parameter λ determines whether merger productivity is closer to the better firm or the worse firm. The curvature parameter θ governs how strongly productivity gains rise with the productivity combination.

3.3 Stand-alone value, merger value, and surplus

The stand-alone value is

$$V(x, z, k) = \pi(P, z, k) + (1 - \delta)k + \max\{U(x, z), 0\}.$$

The value of a merged firm is

$$\begin{aligned} V_M(x, z_i, z_j, k_i, k_j) = {} & \pi(P, z_i, k_i) + \pi(P, z_j, k_j) + (1 - \delta)(k_i + k_j) \\ & + \max\{U(x, \zeta_M(z_i, z_j)), 0\} - c_M. \end{aligned}$$

The merger surplus is

$$W(x, z_i, z_j) = V_M(x, z_i, z_j, k_i, k_j) - V(x, z_i, k_i) - V(x, z_j, k_j).$$

Interpretation: Two firms merge if the surplus is positive. The surplus combines productivity gains, fixed-cost savings, continuation values, and one-time integration costs.

3.4 Merger search

The option value of merging for firm i is

$$W^*(x, z_i) = \int_Z \max\{W(x, z_i, z), 0\} \, dF_M(z; x).$$

If the firm chooses search probability e, its expected net benefit is

$$W(e, x, z_i) = \frac{e}{2} W^*(x, z_i) - C(e).$$

With search cost

$$C(e) = \frac{c_s e}{1 - e},$$

the optimal search effort has the form

$$e^*(x, z_i) = \max\left\{1 - \sqrt{\frac{2c_s}{W^*(x, z_i)}}, 0\right\}.$$

Interpretation: Firms search only when the option value of a potential merger is high enough relative to the search cost. The model therefore produces endogenous merger waves.

3.5 Entry and exit rules

A potential entrant with productivity signal y enters when

$$U(x, y) > c_E.$$

An incumbent exits when

$$U(x, z) < 0.$$

Interpretation: Merger options enter $U(x, z)$. Therefore they can encourage entry and delay exit even before any merger actually occurs.

3.6 Policy-function regressions

The paper summarizes simulated policy functions using regressions of the form

$$Y_t = a + b_s \log s_t + b_N \log N_t + b_m m_t + b_v \log v_t + \varepsilon_t,$$

where Y_t is the merger rate, entry rate, exit rate, or expected output price.

Interpretation: These regressions describe how aggregate demand, industry size, mean productivity, and productivity dispersion affect turnover in the simulated economy.

3.7 Validation regressions

The model's empirical predictions are evaluated using regressions for merger-exit dynamics,

$$ExitRate_{it} = \alpha + \beta_1 MergerRate_{it} + \beta_2 MergerRate_{i,t-1} + \gamma X_{it} + \mu_i + \tau_t + u_{it},$$

cyclicality,

$$MergerCyclicality_i = \alpha + \beta_1 FixedCost_i + \beta_2 VolatilityTFP_i + u_i,$$

and acquirer announcement returns,

$$CAR_i = \alpha + \beta_1 RelativeSize_i + \beta_2 TobinsQ_i + \beta_3 Size_i + u_i.$$

Interpretation: The validation tests ask whether the calibrated model produces empirical regularities not directly targeted in the calibration.

4 Identification and economic design

4.1 What the model identifies

- The model is not a causal reduced-form design. It is a calibrated general-equilibrium experiment.
- The main counterfactual compares the calibrated economy with mergers to an otherwise comparable economy in which the merger implementation cost is set to infinity.
- A second counterfactual lets firms merge but forces entry and exit thresholds to ignore merger options. This separates direct synergy effects from indirect entry-exit effects.

Interpretation: The key object is the difference between equilibrium productivity distributions under alternative institutional environments for mergers.

4.2 Why industry equilibrium matters

- A merger changes the acquiring and target firms directly.
- It also removes one incumbent, changes aggregate output, changes price, and changes the incentive of other firms to enter, exit, or search.
- These equilibrium feedbacks accumulate over time through repeated entry, exit, and mergers.

Interpretation: A partial-equilibrium analysis of realized M&A synergies would miss the indirect entry-exit channel and the effect of mergers on prices and future takeover opportunities.

4.3 Policy counterfactual

The policy exercise varies the entry cost c_E. A policymaker who ignores mergers expects that lower entry costs increase entry and average productivity. The calibrated economy with mergers can imply the opposite because lower entry costs reduce prices, reduce the gains from mergers, and lower the merger rate.

Interpretation: The paper's most important policy lesson is that entry-promoting policies may have different productivity effects once the merger market is part of the equilibrium system.

5 Tables and figures

5.1 Figure 1 + Figure 2: timing and merger productivity

Read:

- Figure 1 shows the order of decisions: incumbents choose search effort, merge if matched, invest, receive payoff, and choose exit; entrants observe signals and decide whether to pay the entry cost.
- Figure 2 shows how the productivity of the merged firm depends on the two merging firms' productivities.
- When $\lambda = 1$, the merged firm's productivity is closer to the more productive firm.
- When $\lambda = 0$, the merged firm's productivity depends more on the less productive firm.

Economic message: The model treats mergers as endogenous reallocations that affect future productivity, not as one-period transactions only.

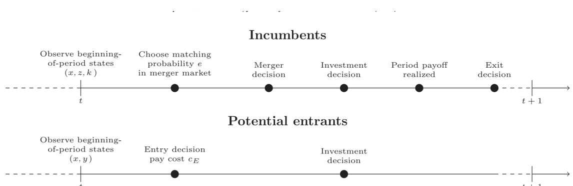
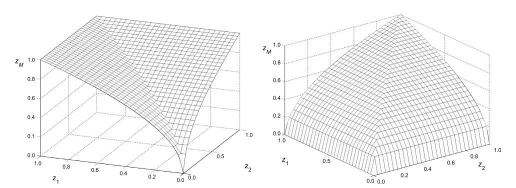


Fig. 1. Timing of events within each period for incumbents and potential entrants. x denotes the vector of aggregate states, z the idiosyncratic productivity shock, k firm's capital, and y the productivity signal observed by potential entrants.

(a) Timing of events



(a) $\lambda = 1$ (b) $\lambda = 0$

Fig. 2. Productivity of the merged firm, ζ_M, as a function of the productivities of the two merging firms, z_1 and z_2. In both figures, $\theta = \frac{1}{2}$.

(b) Productivity of merged firm

5.2 Tables 1–3: parameters, calibration targets, and model variables

Read:

- Table 1 reports standard parameters and calibrated parameters. The calibrated merger implementation cost is 0.313, search-cost parameter is 0.023, synergy weight is 0.620, and entry cost is 0.291.
- Table 2 shows that the model fits the average merger rate, autocorrelation of merger rate, combined gains, exit rate, and aggregate earnings moments reasonably well.
- Table 3 maps model objects to empirical moments such as Tobin's Q, combined merger gains, exit rate, merger rate, entry rate, and matching rate.

Economic message: The calibration discipline is broad: the model is forced to match both merger-market moments and entry/exit moments.

Table 1
Parameter values.
Panel A reports the set of parameters that are chosen based on standard values in the literature. Panel B displays the values of the parameters that are calibrated to match the empirical moments in Table 2.

Symbol	Description	Value
Panel A: Standard parameters		
α	Returns-to-scale parameter	0.750
δ	Depreciation of capital	0.150
ρ_z	Persistence of idiosyncratic productivity shock	0.660
σ_z	Standard deviation of idiosyncratic productivity shock	0.110
r	Discount rate	0.040
Panel B: Calibrated parameters		
c_M	Fixed merger-implementation cost	0.313
c_s	Search-cost parameter	0.023
λ	Most-productive-firm synergy weight	0.620
θ	Curvature parameter of the synergy function	0.490
c_f	Fixed cost of production	0.154
c_E	Cost of entry	0.291
ρ_s	Persistence of aggregate demand shock	0.609
σ_s	Standard deviation of aggregate demand shock	0.069
η	Price elasticity of demand	2.173
N_E	Mass of potential entrants	0.962

(a) Parameter values

Table 2
Calibration targets.
The table contains the empirical moments targeted in the calibration and their simulation counterparts. Column 1 presents the moments simulated from the model using the parameter values in Table 1. The definitions of the variables in the model used to compute the moments are in Table 3. Column 2 shows the value of the empirical moments from the data. The definition of variables and the details about sample construction are provided in Appendix C. The merger and exit rates are computed at an annual frequency using data from the Center for Research in Security Prices (CRSP). Information on Tobin's Q for merged firms is from Thomson Reuters' SDC Platinum and the CRSP/Compustat merged database. Aggregate earnings growth is measured using the time series of aggregate net operating surplus for nonfinancial domestic corporate business from the Bureau of Economic Analysis. The estimates of average combined merger gains are from Betton, Eckbo and Thorburn (2008), and total factor productivity data are from İmrohoroğlu and Tüzel (2014).

Description of moment	Model (1)	Data (2)
Average merger rate	4.21%	4.54%
Standard deviation of merger rate	0.012	0.016
Autocorrelation of merger rate	0.292	0.291
Average absolute scaled difference in log Tobin's Q of merged firms	0.925	0.953
Average combined merger gains	1.63%	1.79%
Average exit rate	3.71%	3.68%
Average relative total factor productivity of entrants	1.060	1.229
Average relative total factor productivity of exiting firms	0.722	0.791
Autocorrelation of aggregate earnings growth	0.182	0.156
Standard deviation of aggregate earnings growth	0.093	0.088

We calibrate the ten parameters in Panel B of Table 1 to

(b) Calibration targets

Table 3
Definition of variables in the model.
x denotes the vector of aggregate states, z the productivity shock, k firm's capital, N the mass of incumbents, N_E the mass of potential entrants, $F(z)$ the start-of-period cross-sectional distribution of firm productivities. $J(x, z, k)$ is the value of the firm at the start of the period, defined by Eq. (11). $W(x, z_i, z_j)$ are the synergies generated by a merger between firm i and j, defined in Eq. (5). $W^*(x, z_i)$ is the option value of merging for firm i, defined in Eq. (6). $e^*(x, z_i)$ is the optimal search effort for firm i, defined in Eq. (8). $P_E(x)$, $P_\epsilon(x)$, $P_M(x)$, and $P_N(x)$ are probabilities defined by Eqs. (17)–(20), respectively. The remaining parameters are defined in Table 1. In the numerical solution, the vector of aggregate states x is approximated by $\widehat{x} \equiv [s, P, \log(N), m, \log(v)]$, where P is output price, and m and v are, respectively, the mean and variance of log-productivity at the start of each period.

Variable	Definition
Tobin's Q	$\frac{J(x,z,k)}{k}$
Combined merger gains for firms i and j	$\frac{W(x,z_i,z_j)-\frac{1}{2}(e^*(x,z_i)W^*(x,z_i)+e^*(x,z_j)W^*(x,z_j))}{J(x,z_i,k_i)+J(x,z_j,k_j)}$
Exit rate	$\left(1-\int_{\mathcal{Z}} e^*(x,z)\,\mathrm{d}F(z)\right)(1-P_\epsilon(x))+\int_{\mathcal{Z}} e^*(x,z)\,\mathrm{d}F(z)\left(1-\frac{P_M(x)}{2}-P_N(x)\right)$
Merger rate	$\frac{P_M(x)}{2}$
Entry rate	$\frac{P_E(x)N_E}{N}$
Matching rate	$\int_{\mathcal{Z}} e^*(x,z)\,\mathrm{d}F(z)$

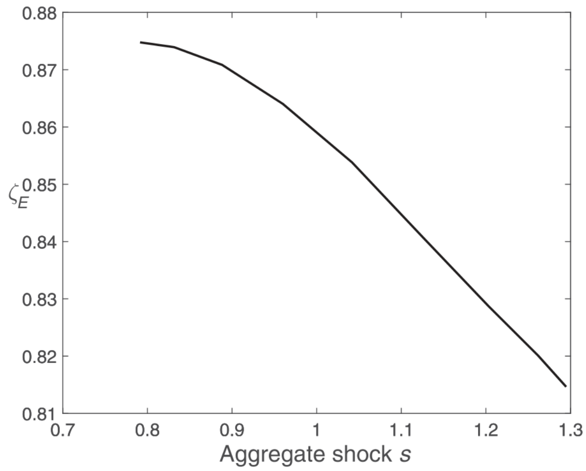
Definitions of model variables and empirical counterparts

5.3 Figure 3 + Figures 4–6: policy functions and merger sets

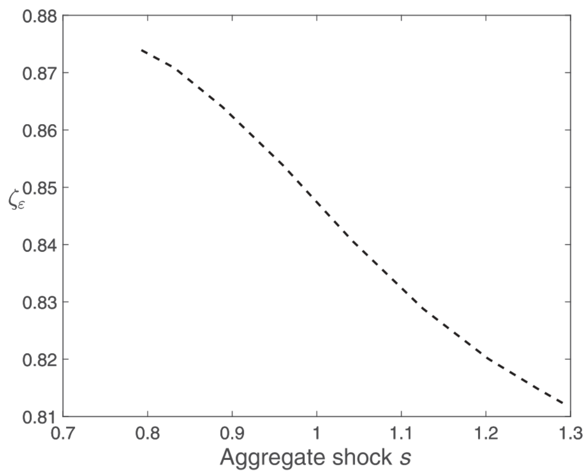
Read:

- Figure 3 shows that entry and exit productivity thresholds decline with aggregate demand. Entry is procyclical and exit is countercyclical.
- Figure 4 shows that merger synergies are hump-shaped in the productivities of the two merging firms.
- Figure 5 shows the merger opportunity set, search set, and equilibrium search effort. Not all firms with possible gains search because search is costly.
- Figure 6 shows that merger synergies are small or negative at low aggregate demand but become positive for a wider set of firm pairs when demand is high.

Economic message: Merger activity is endogenous. It responds to demand, price, productivity dispersion, and the distribution of potential partners.



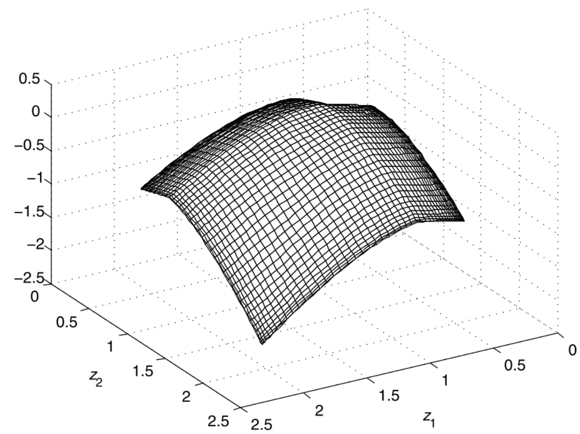
(a) Entry threshold (ζ_E)



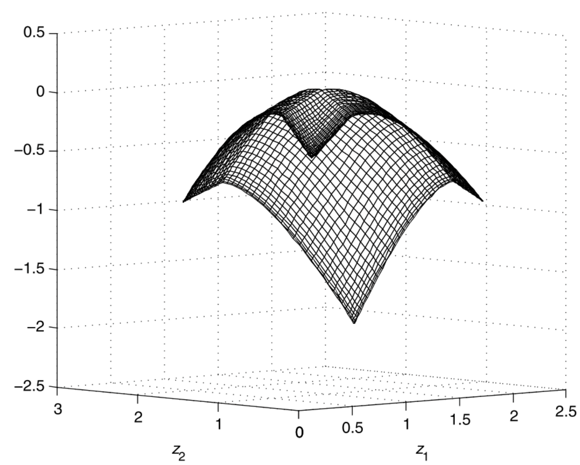
(b) Exit threshold (ζ_ϵ)

Fig. 3. Entry and exit productivity thresholds (ζ_E, ζ_ϵ) as a function of the aggregate demand shock (s). The data are generated according to the algorithm described in the online appendix, and the parameter values are shown in Table 1. States m, v, and N are set at their average levels in the simulation.

(a) Entry and exit thresholds

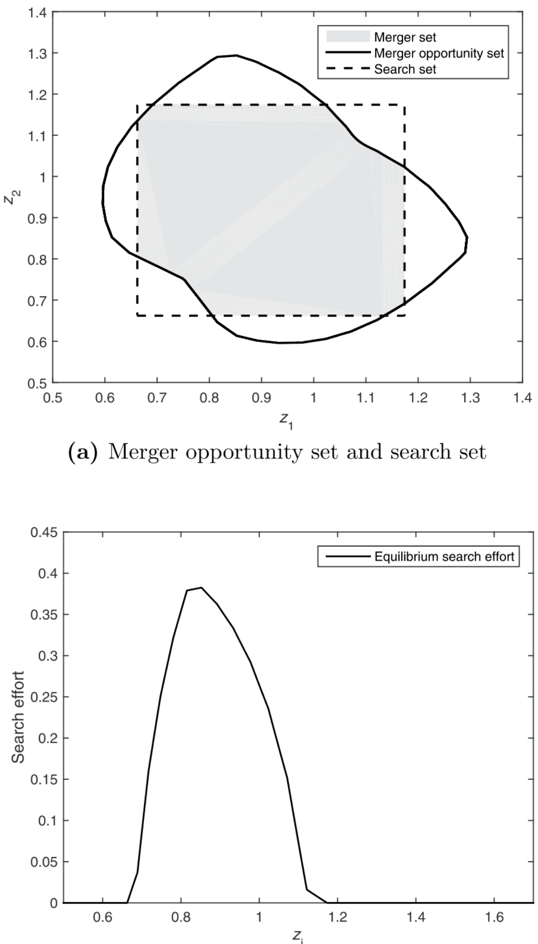


(a) Merger synergies: Angle 1



(b) Merger synergies: Angle 2

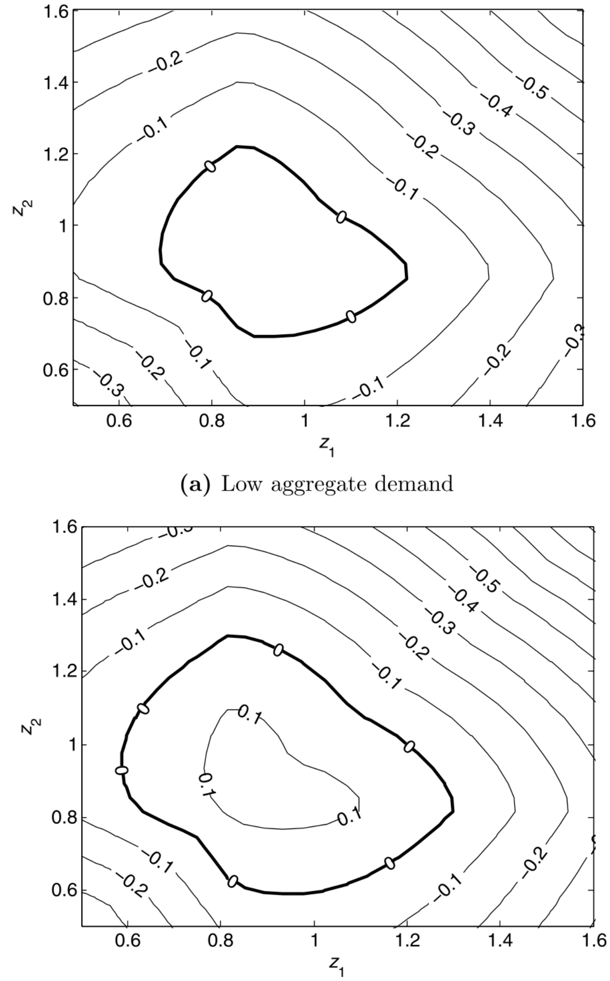
(b) Merger synergy surface



(a) Merger opportunity set and search set

(b) Search effort

Fig. 5. Merger synergies and search effort. Panel (a) shows the merger



(a) Low aggregate demand

(b) High aggregate demand

(a) Merger opportunity and search set

(b) Synergies and aggregate demand

5.4 Tables 4–5 and Figures 7–8: effects of merger options

Read:

- Table 4 shows policy-function regressions. With mergers, the merger rate rises with aggregate demand and productivity dispersion, and falls with mean productivity and the mass of incumbents.
- The presence of mergers raises entry and reduces exit relative to the no-merger economy.
- Table 5 shows that average productivity is 1.1059 with mergers and 1.0556 without mergers.
- The total productivity effect of mergers is 0.0476, the direct effect is 0.0409, and the indirect entry-exit effect is 0.0068.
- Figure 7 shows that productivity gains and merger rates vary strongly with the synergy-weight parameter, the curvature parameter, and search costs.
- Figure 8 shows that search incentives vary by firm productivity and by the shape of the merger productivity function.

Economic message: Most of the productivity gain comes from realized synergies, but the option to merge also changes the composition of firms in the industry.

Table 4

Regression analysis of the policy functions.

This table presents ordinary least squares (OLS) coefficients for regressions of the simulated entry, exit, and merger rates and the expected price on the aggregate state variables. The entry, exit, and merger rates and expected price are computed in the simulation step of the solution algorithm described in the online appendix. Definitions of the entry, exit, and merger rates are provided in Table 3. The expected price is defined in Eq. (16). For Columns 1 to 4, the parameters used to simulate the economy are set to the values shown in Table 1. In Columns 5 to 7, we simulate an economy without mergers by setting the merger-implementation cost, c_M, to infinity. s denotes the aggregate demand shock. m and v are, respectively, the mean and variance of the cross-sectional distribution of firms' log-productivity shocks, z. N denotes the mass of incumbents. Standard errors of the ordinary least squares regression coefficients are reported in parentheses.

		With mergers			Without mergers		
Variable	Merger rate (1)	Entry rate (2)	Exit rate (3)	Expected price (4)	Entry rate (5)	Exit rate (6)	Expected price (7)
$log(s)$	0.1546	0.2924	−0.1312	0.0645	0.2638	−0.1524	0.0604
	(0.0028)	(0.0016)	(0.0019)	(0.0002)	(0.0017)	(0.0008)	(0.0002)
$log(N)$	−0.1131	−0.2880	0.1228	−0.0432	−0.2873	0.1839	−0.0382
	(0.0086)	(0.0050)	(0.0056)	(0.0005)	(0.0055)	(0.0025)	(0.0006)
m	−0.4349	−0.5657	−0.1845	−0.0494	−0.2332	−0.2789	−0.0303
	(0.0533)	(0.0308)	(0.0351)	(0.0033)	(0.2237)	(0.1010)	(0.0224)
$log(v)$	0.0279	0.0232	0.0699	0.0030	0.0844	0.1065	0.0067
	(0.0147)	(0.0085)	(0.0097)	(0.0009)	(0.0286)	(0.0129)	(0.0029)
Constant	0.2661	0.4932	0.2475	0.3545	0.7346	0.3284	0.3620
	(0.0595)	(0.0343)	(0.0392)	(0.0037)	(0.0994)	(0.0449)	(0.0100)
Adjusted R^2	0.8972	0.9894	0.9392	0.9976	0.9867	0.9912	0.9975
Observations	375	375	375	375	375	375	375

Regression analysis of policy functions

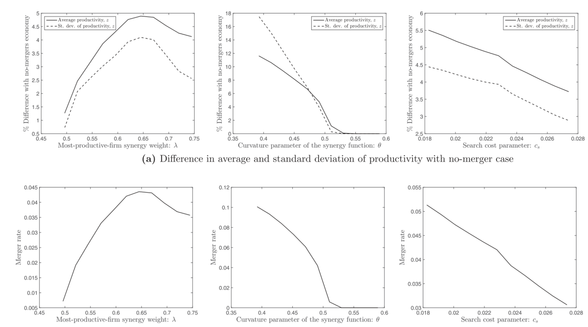
Table 5

Effect of merger options on productivity.

This table provides the results of counterfactual experiments that show the impact of merger options on average firm productivity as a function of the structural parameters. We present three sets of results, corresponding to an economy that allows for merger activity ("With mergers"), an economy in which firms cannot merge ("Without mergers"), and a third one in which firms can merge, but ignore future merger options when making their entry and exit decisions ("Ignoring M&A options"). Column 1 presents the results for the main calibration (parameter values in Table 1). The remaining columns show the results corresponding to a 1% increase from each parameter's calibrated value. The merger rate is defined in Table 3. We compute mean productivity taking the time-series average of the cross-sectional mean firm productivity, z. "Total effect" is the change in mean productivity in the merger economy compared to the economy without mergers. The "Direct effect" is the change in mean productivity in the economy ignoring merger options compared to the economy without mergers. The "Indirect effect" is the difference between the total and direct effects.

Variable	Type of economy	Base case (1)	λ (2)	θ (3)	c_M (4)	c_s (5)	c_f (6)	c_E (7)	N_E (8)	ρ_s (9)	η (10)	σ_s (11)
Av. merger rate	With mergers	0.0421	0.0428	0.0348	0.0386	0.0415	0.0446	0.0428	0.0417	0.0431	0.0424	0.0421
Mean product.	With mergers	1.1059	1.1064	1.0983	1.1024	1.1053	1.1085	1.1063	1.1056	1.1069	1.1055	1.1058
	Ignoring M&A opt.	1.0987	1.1028	1.0785	1.0922	1.0992	1.1031	1.0992	1.1006	1.1018	1.1023	1.1018
	Without mergers	1.0556	1.0556	1.0556	1.0556	1.0556	1.0556	1.0556	1.0557	1.0556	1.0553	1.0557
Total effect	With/Without (a)	0.0476	0.0481	0.0404	0.0443	0.0471	0.0501	0.0482	0.0472	0.0486	0.0476	0.0475
Direct effect	Ignor./Without (b)	0.0409	0.0447	0.0217	0.0347	0.0413	0.0450	0.0461	0.0425	0.0437	0.0446	0.0437
Indirect effect	Difference (a-b)	0.0068	0.0034	0.0187	0.0096	0.0058	0.0051	0.0020	0.0047	0.0048	0.0030	0.0038

(a) Effect of merger options on productivity

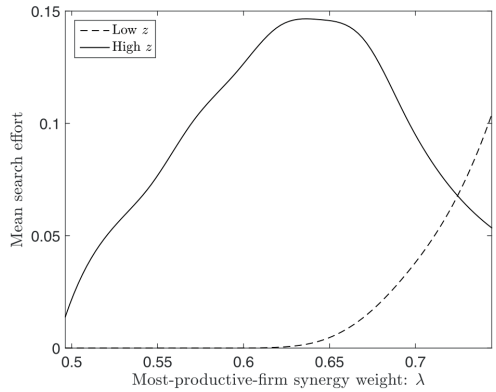


(a) Difference in average and standard deviation of productivity with no-merger case

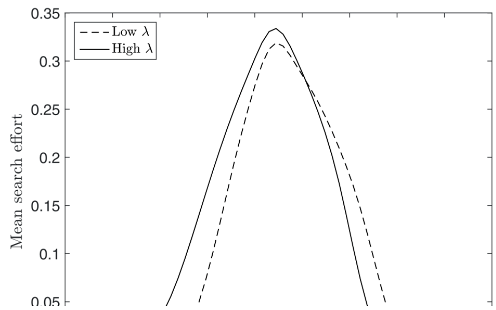
(b) Average merger rate

Fig. 7. Mergers and productive efficiency. The three figures in Panel (a) show the percentage difference in the mean (solid line) and standard deviation (dashed line) of firm productivity z between economies with and without mergers, as a function of the most-productive-firm synergy weight, λ, the curvature parameter of the synergy function, θ, and the search cost, c_s, respectively. Panel (b) shows the average merger rate as a function of λ, θ, and c_s, respectively. The other parameter values are shown in Table 1. The economy without mergers is obtained by setting the merger-implementation cost, c_M,

(b) Mergers and productive efficiency



(a) Mean search effort as a function of the most-productive-firm synergy weight λ



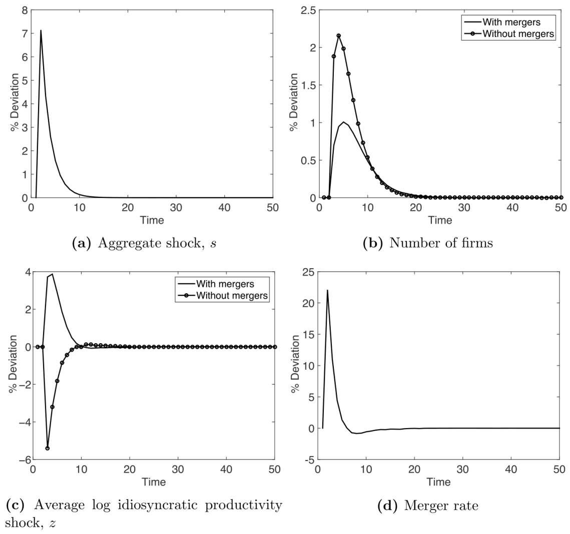
Search intensity

5.5 Figures 9–12: business cycles and entry-cost policy

Read:

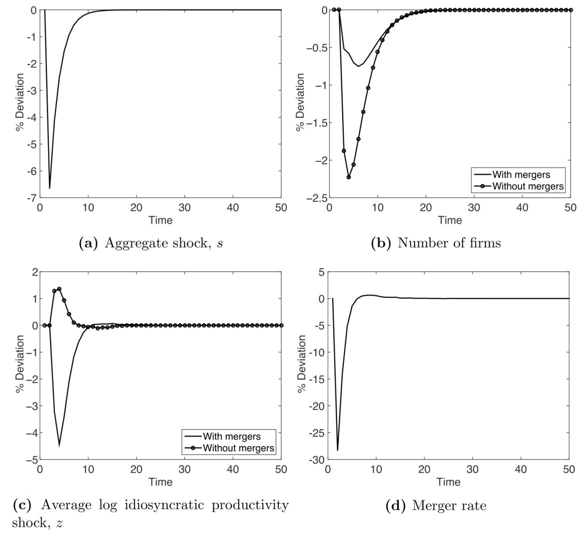
- Figure 9 shows a positive demand shock. Without mergers, average productivity falls because low-productivity firms enter or remain. With mergers, average productivity rises because merger synergies create a merger wave.
- Figure 10 shows a negative demand shock. Without mergers, average productivity rises because exit is cleansing. With mergers, productivity falls because merger synergies collapse and fewer low-productivity firms are acquired.
- Figure 11 shows what a policymaker expects if she ignores mergers: lowering entry costs raises entry and raises average productivity.
- Figure 12 shows what happens when merger activity is correctly included: lowering entry costs can lower productivity because the merger rate and merger synergies decline.

Economic message: The model reverses simple policy intuition. Entry-promotion can reduce average productivity if it weakens the merger market enough.



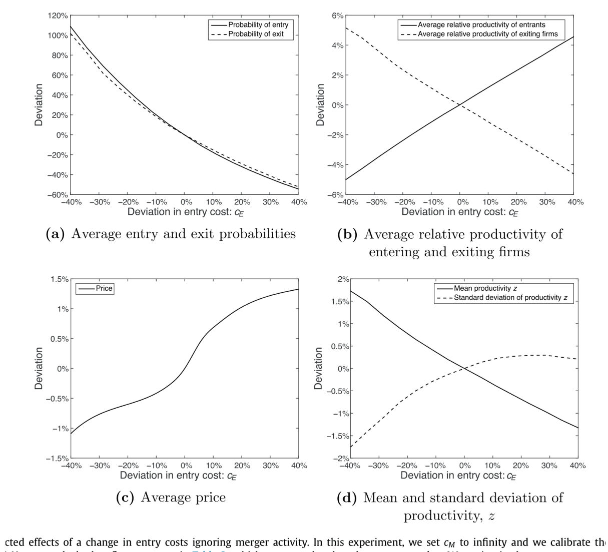
(a) Aggregate shock, s (b) Number of firms (c) Average log idiosyncratic productivity shock, z (d) Merger rate

(a) Positive demand shock



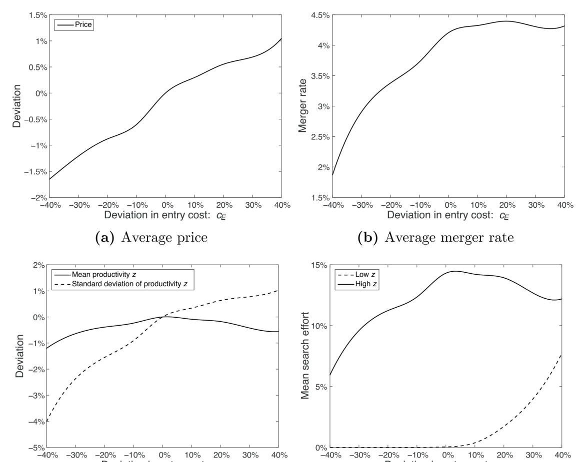
(a) Aggregate shock, s (b) Number of firms (c) Average log idiosyncratic productivity shock, z (d) Merger rate

(b) Negative demand shock



(a) Average entry and exit probabilities (b) Average relative productivity of entering and exiting firms (c) Average price (d) Mean and standard deviation of productivity, z

(a) Entry costs ignoring M&A



(a) Average price (b) Average merger rate

(b) Entry costs accounting for M&A

5.6 Tables 6–9 and Figure 13: robustness and validation

Read:

- Table 6 shows that the main conclusions survive alternative exit rates, time-varying discount rates, and higher entry costs.
- Table 7 validates the predicted merger-exit dynamics. In both model and data, merger activity is negatively related to exit, including lagged exit.
- Figure 13 shows that merger cyclicality depends on synergy sources. The merger rate becomes less procyclical when fixed-cost savings or productivity-shock volatility matter more.
- Table 8 shows that industries with higher fixed operating costs and higher volatility of TFP shocks have lower merger cyclicality.

- Table 9 shows that acquirer returns decline with relative target size and acquirer size, aligning the model with well-known empirical M&A return patterns.

Economic message: The validation exercises support the central mechanism: mergers are tied to exit dynamics, business-cycle conditions, and the distribution of firm productivity.

Table 6
Robustness checks.

This table shows the results of three robustness checks, in which we recalibrate the parameters of the model and perform counterfactual experiments. In the first robustness check (Panel A), the target exit rate in the calibration is set to 2%. In the second (Panel B), firms discount future cash flows using the pricing kernel in Eq. (33). In the third (Panel C), we set the cost of entry, c_E, to 0.582. For each robustness check, we present two sets of results, corresponding to a calibration that accounts for merger activity ("With mergers") or ignores it ("Without mergers"). In the calibrations without mergers, we set c_E at the value in the corresponding economy with mergers, and we match the five merger-unrelated moments in Table 2. The lines "$c_M \to \infty$" and "c_E reduced by 40%" show comparative statics results that obtain from setting c_M to infinity and reducing c_E by 40%, respectively. Column 1 shows the time-series mean of average productivity z in the cross-section. Column 2 shows the mean value of the cross-sectional standard deviation of productivity. Column 3 shows the time-series correlation of mean cross-sectional productivity and the aggregate demand shock $\log(s)$. Columns 4 and 5 show the mean entry and exit rates, respectively.

Type of economy	Robustness check	Mean productivity (1)	St. dev. of productivity (2)	Cyclicality of mean productivity (3)	Mean entry rate (4)	Mean exit rate (5)
Panel A: Exit rate						
With mergers	At calibrated parameters	1.1109	0.1576	0.5148	0.0746	0.0247
	$c_M \to \infty$	1.0512	0.1497	−0.3379	0.0524	0.0521
	c_E reduced by 40%	1.1165	0.1553	0.5102	0.106	0.0576
Without mergers	At calibrated parameters	1.0335	0.1506	−0.4102	0.0296	0.0216
	c_E reduced by 40%	1.0487	0.1478	−0.4595	0.0539	0.0537
Panel B: Discount rate						
With mergers	At calibrated parameters	1.1214	0.1567	0.3867	0.0934	0.041
	$c_M \to \infty$	1.064	0.1527	−0.0071	0.0636	0.0628
	c_E reduced by 40%	1.1251	0.1527	0.4086	0.1376	0.0956
Without mergers	At calibrated parameters	1.0463	0.152	0.1924	0.039	0.0385
	c_E reduced by 40%	1.0676	0.1494	−0.1725	0.08	0.079
Panel C: Entry cost						
With mergers	At calibrated parameters	1.1126	0.1561	0.3439	0.0802	0.0388
	$c_M \to \infty$	1.0687	0.1525	−0.3441	0.0727	0.0722
	c_E reduced by 40%	1.0918	0.1483	−0.0165	0.1221	0.1201
Without mergers	At calibrated parameters	1.044	0.1525	−0.1323	0.0347	0.0345
	c_E reduced by 40%	1.066	0.1501	−0.3983	0.0751	0.0746

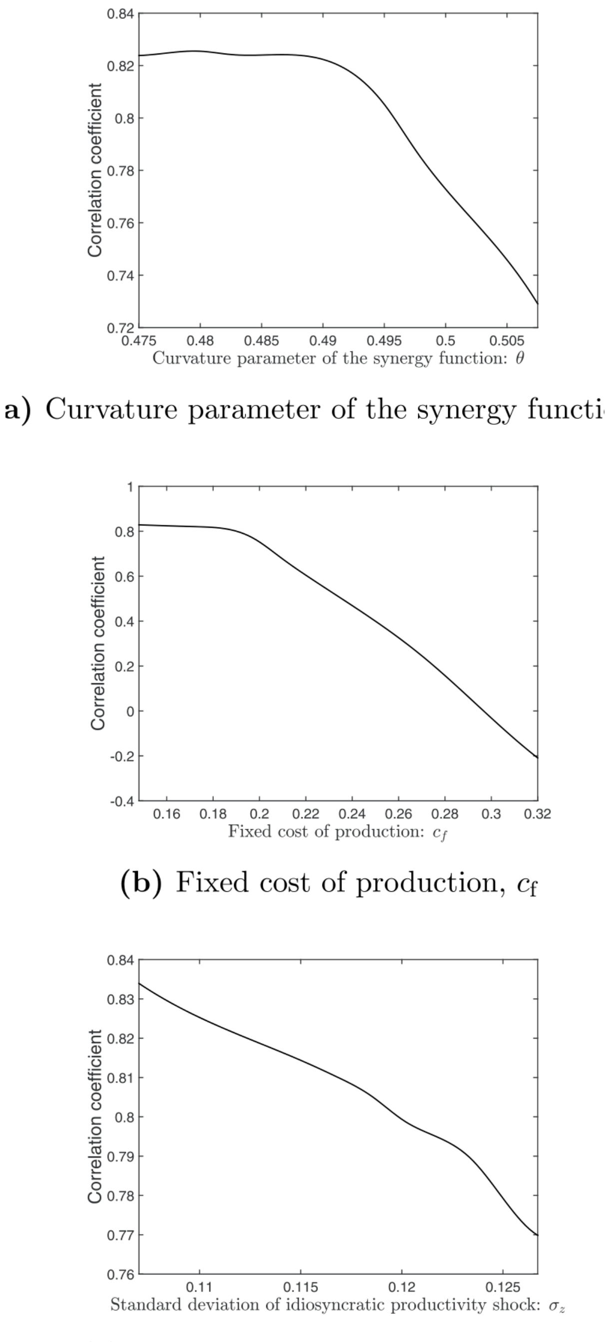
(a) Robustness checks

Table 7
Merger- and exit-rate regressions.

This table presents coefficient estimates for merger- and exit-rate regressions. Columns 1 and 2 present the OLS coefficient estimates for the simulated data generated according to the parameters in Table 1 and following the algorithm described in the online appendix. Definitions of the entry, exit, and merger rates are provided in Table 3. s denotes the aggregate demand shock. m and v are, respectively, the mean and variance of the cross-sectional distribution of firms' log-productivity shocks, z. P denotes the product price. Columns 3 to 6 present the estimates obtained using the real data. The definition of the variables and the construction of the sample are described in Appendix C. The merger and exit rates are computed at an annual frequency using data from CRSP for the period 1981–2010. *Profitability* is the average return on assets for the firms in the same industry and year. Industries are defined using the Fama and French 49-industry classifications. In Columns 4 and 6, we use the dynamic panel data estimator of Blundell and Bond (1998). Robust standard errors are reported in parentheses.

	Model		Data			
Variable	Exit rate (1)	Merger rate (2)	Exit rate (3)	Exit rate (4)	Merger rate (5)	Merger rate (6)
Merger rate	−0.573	–	−0.136	−0.143	–	–
	(0.0201)		(0.0293)	(0.0326)		
Lagged merger rate	−0.0500	−0.276	−0.0775	−0.0823	–	−0.0439
	(0.0250)	(0.0624)	(0.0313)	(0.0356)		(0.0410)
Lagged exit rate	0.0182	−0.188	–	0.0840	−0.0959	−0.0919
	(0.0355)	(0.109)		(0.0614)	(0.0305)	(0.0286)
Profitability	–	–	−0.0739	−0.0586	−0.00837	−0.00800
			(0.0421)	(0.0350)	(0.0215)	(0.0236)
$\log(s)$	−0.00339	0.0792	–	–	–	–
	(0.0128)	(0.0426)				
$\log(P)$	−0.0398	0.0769	–	–	–	–
	(0.0126)	(0.0438)				
$\log(N)$	0.0345	−0.0707	–	–	–	–
	(0.00826)	(0.0261)				
m	−0.452	−0.429	–	–	–	–
	(0.0399)	(0.107)				
$\log(v)$	0.0819	0.0440	–	–	–	–
	(0.00810)	(0.0193)				
Constant	0.356	0.399	–	–	–	–
	(0.0342)	(0.0888)				
Year fixed effects	–	–	Yes	Yes	Yes	Yes
Industry fixed effects	–	–	Yes	Yes	Yes	Yes
Adjusted R^2	0.9860	0.8980	0.2033	–	0.1845	–
Observations	374	374	1,247	1,247	1,247	1,247

(b) Merger- and exit-rate regressions



(a) Determinants of merger cyclicality

Table 8
Determinants of merger cyclicality.

This table presents ordinary least squares estimates for regressions of merger cyclicality on industry-level variables. The dependent variable, merger cyclicality, is computed at the industry level as the coefficient of a time-series regression of the annual merger rate on real US GDP growth and a constant. The variable *Fixed operating costs* is the constant coefficient estimated from industry-level regressions of the firm-specific operating-costs-to-assets ratio on the sales-to-assets ratio. The variable *Volatility of shocks to TFP* is computed, for each industry, as the standard deviation of the residuals from a regression of log firm TFP on its lagged value and a constant. We use the total factor productivity estimates provided by İmrohoroğlu and Tüzel (2014). Industries are defined using the Fama and French 49-industry classifications, and we exclude utilities and financial companies. Appendix C contains the details of sample construction and variable definitions. Robust standard errors are reported in parentheses.

Variable	Merger cyclicality (1)	Merger cyclicality (2)	Merger cyclicality (3)
Fixed operating costs	−1.3606	–	−0.9644
	(0.2709)		(0.3276)
Volatility of shocks to TFP	–	−1.6728	−1.1148
		(0.4370)	(0.5233)
Constant	0.2901	0.7720	0.6196
	(0.0406)	(0.1268)	(0.1519)
R^2	0.250	0.233	0.333
Observations	43	43	43

merger rate with gross-domestic-product (GDP) growth, is
higher in industries with low fixed operating costs (Col-

(b) Regression evidence on merger cyclicality

Table 9
Regression analysis of acquirer returns.

This table presents ordinary least squares estimates for regressions of acquirer returns on acquirer and target characteristics. The regressions are performed using data simulated according to the procedure described in Section 5.3. We assume that, in each simulated deal, the acquirer is the firm with the highest preacquisition market value, $J(x, z, k)$, defined in Eq. (11). *Relative size* is defined as the ratio between the deal value and the market value of the acquirer. *Tobin's Q* is defined in Table 3. *Small acquirer* is an indicator variable equal to one for firms in the bottom quintile of preacquisition market value in a given simulation period. *Log of acquirer assets* is measured as the log of the acquirer's stock of capital at the start of the period. Standard errors of the ordinary least squares regression coefficients are reported in parentheses.

Variable	Acquirer return (1)	Acquirer return (2)	Acquirer return (3)	Acquirer return (4)	Acquirer return (5)
Relative size	−0.0137	−0.0142	−0.0142	0.0013	−0.0186
	(0.0002)	(0.0002)	(0.0002)	(0.0004)	(0.0002)
Acquirer Tobin's Q	−0.1302	0.1301	0.0894	−0.0305	−0.1958
	(0.0023)	(0.0108)	(0.0089)	(0.0034)	(0.0011)
Small acquirer	0.0006	–	–	–	–
	(0.0001)				
Log of acquirer market value	–	−0.0464	–	–	–
		(0.0018)			
Log of acquirer assets	–	–	−0.0468	–	–
			(0.0017)		
Constant	0.1722	−0.0802	−0.0396	0.0523	0.2510
	(0.0027)	(0.0106)	(0.0088)	(0.0039)	(0.0014)
Adjusted R^2	0.727	0.748	0.750	0.292	0.854
Observations	20084	20084	20084	5812	14272

6 Short summary

- The paper studies mergers in dynamic industry equilibrium with heterogeneous firms.
- Mergers improve productivity directly through synergies and indirectly through their effect on entry and exit.
- The model is calibrated to merger rates, merger gains, exit rates, productivity of entrants and exits, and aggregate earnings moments.
- In the calibrated model, mergers raise average productivity by 4.8% relative to no mergers.
- Direct synergy accumulation explains about 4.1%; entry-exit effects explain about 0.7%.
- Merger options raise entry because new firms value the possibility of being acquired.
- Merger options reduce exit because poor performers wait for potential acquisition opportunities.
- Merger activity makes productivity more procyclical in booms because synergies become more valuable.
- Recessions can become sullying when mergers are present because merger activity and synergies fall.
- Policies that reduce entry costs can lower productivity if they weaken the merger market.
- The paper validates the model using merger-exit regressions, merger-cyclicality regressions, and acquirer-return regressions.

7 Conclusion

7.1 Core conclusion

Dimopoulos and Sacchetto show that M&A activity is not merely a direct transfer of assets between firms. In industry equilibrium, mergers reshape entry, exit, prices, and the distribution of productivity. The calibrated model implies that mergers raise average productivity materially, with most of the gain coming from realized synergies and a smaller but important component coming from altered entry and exit decisions.

7.2 Economic intuition

The intuition is that a merger option is a real option. It makes entry more attractive because entrants may later be acquired. It makes exit less attractive because weak incumbents may be rescued through acquisition. It also changes how aggregate shocks affect productivity. In booms, productivity-improving mergers become more valuable and create merger waves. In downturns, those synergies shrink, and mergers no longer clean up low-productivity firms.

7.3 Takeaway

The paper's main lesson is that merger policy and entry policy cannot be evaluated separately. An active takeover market can improve productivity, but it also changes the behavior of entrants and weak incumbents. Public policies that promote entry may have unexpected productivity effects if they alter prices and reduce the incentives for productivity-enhancing mergers.